

POTENTIAL CHALLENGES / RISKS

1

2D Simulation Latency Bottleneck

Solving 2D Saint-Venant shallow water equations across 400 km² takes 3–5 hours, violating the 0–3 hr nowcasting window.

2

Drain Siltation & Clogging

Urban catch-pits choke with solid waste and silt, causing violent manhole surcharge not accounted for in clean CAD blueprints.

3

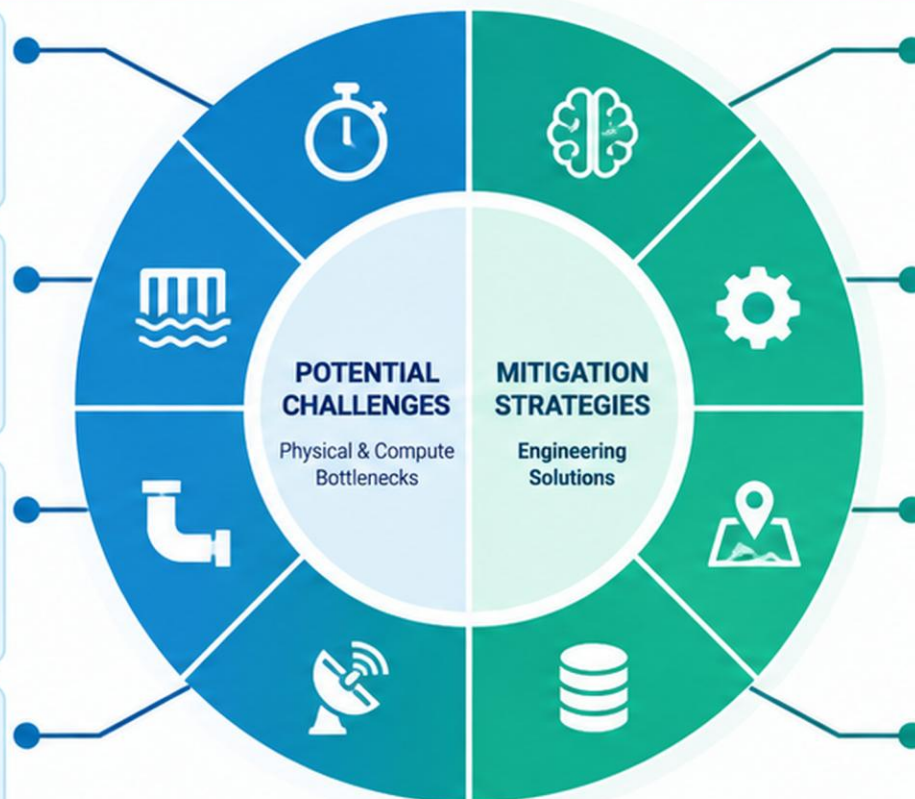
Unmapped Underground Drains

Fragmented or legacy paper blueprints in municipal wards leave spatial blind spots in subsurface pipe network topology.

4

Radar Attenuation & Data Gaps

Doppler radar suffers ground clutter and beam attenuation; gauges only record rain after it lands at isolated points.



MITIGATION STRATEGIES

1

Physics-Informed GNN (< 350 ms)

Pre-trained offline on 2D hydrodynamic runs; executes sub-second spatial inference during live storms for true 0–3 hr lead time.

2

Dynamic Clogging Factor (μ_{clog})

Ingests municipal desilting logs, zonal MSW tonnage (TPD), and 1913 complaints to dynamically throttle pipe intake capacity.

3

Topographic Flow Inversion

Combines OSM street centerlines with DEM flow-accumulation paths to synthesize directed drainage graphs for unmapped wards.

4

Multi-Source Sensor Fusion

Adaptive Kalman filtering using IMD Doppler radar (10-min scans) with NASA GPM DPR satellite data & municipal AWS telemetry.

RESULTING FEASIBILITY & VIABILITY



1. TECHNICAL FEASIBILITY

- Zero sensor capex
- Uses open IMD radar & Cartosat DEM
- Proven models + modern ML



2. OPERATIONAL VIABILITY

- 0–3h lead time, cm depth resolution
- Safe A* navigation API
- Integration with existing systems



3. ECONOMIC VIABILITY

- Cloud-native microservice (NIC/AWS)
- Saves 100s Cr in flood loss
- Minimal additional hardware



4. PAN-INDIA SCALABILITY

- Calibrated on Chennai
- 100% portable to Mumbai, Delhi, Bengaluru
- Modular and configurable